

Ayman Kamel Korany Bayoumi

Scientific Computing and Research Software

Computational Mechanics | Numerical Modeling | Scientific AI Evaluation

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ORCID: 0009-0005-0592-408X | Scientific profile and Kepler work

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Professional Summary

Mechanical Engineering and Computer Science graduate with hands-on Python experience in numerical modeling, scientific software and AI evaluation. Current Kepler project work combines coupled thermal and kinetic models, independent solver comparisons, command-line workflows and automated verification. Background includes nonlinear FEM, structural benchmark authoring and production software. Focused on research engineering that turns explicit scientific models into reproducible, objectively checkable computations.

Technical Skills

Scientific computing: Python, NumPy, SciPy, ODE integration, numerical linear algebra, optimization, heat transfer, FEM and modal analysis.

Verification: pytest, Hypothesis, conservation checks, convergence studies, reference comparisons, numerical tolerances and scientific AI evaluation.

Workflow tools: Git, structured JSON, command-line interfaces, PowerShell, Linux tooling, Docker and CI workflows. Recent Kepler scientific runs are documented on Windows/PowerShell.

Kepler Project Work

Computational Physics and Scientific AI Evaluation | 2026 - Present

- Developed and reviewed a reduced Al-Zr materials model coupling slab heat conduction to integer-cluster kinetics, with thermal and chemical history retained across processing and service.
- Worked with positive backward-Euler and independent BDF solvers, command-line evaluation and bounded optimization of furnace programs using Python, NumPy and SciPy.
- Implemented and checked structured inputs, unit validation, conservation and detailed balance, selected thermal-grid and cluster-size refinements, input hashes, diagnostics and execution journals.
- Evaluated AI-generated solutions to generalized eigenvalue problems with repeated modes using projected perturbation matrices, directional derivatives, finite-difference comparisons and eigenspace basis rotations.
- Current applications use synthetic inputs in materials physics and mechanics; heat transport and numerical validation provide methods relevant to geophysical and environmental modeling.

Scientific Benchmark Experience

Numerical Solver / Structures Task Contributor | Prometheus | 2026

- Developed structural-engineering benchmark tasks with problem definitions, reference solutions, automated tests, scoring logic and reproducible evaluation workflows.
- Worked with 2D frames, FEM, P-Delta effects, geometric stiffness, static condensation, force recovery and stability checks; applied numerical tolerances and reference comparisons.

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Selected Scientific and Software Projects

coroFEM | Nonlinear 2D Frame Finite Element Solver | Python | 2026

- Developed a NumPy/SciPy Euler-Bernoulli and corotational frame solver for linear statics, geometric nonlinearity and linearized eigenvalue buckling.
- Implemented Newton-Raphson iteration, displacement control and Crisfield arc-length continuation; built analytical verification cases and OpenSeesPy comparisons.

ModelMesh | Constraint-Satisfaction Solver | Python | 2026

- Implemented AC-3/AC-4, global constraints and MRV, dom/wdeg and LCV search heuristics for scheduling and classic constraint problems, with Hypothesis property-based checks.

Railway Secretariat System | Flutter and Dart | 2025 - 2026

- Built correspondence-management software with SQLite, a Dart backend, OCR/PDF import, authentication hardening, automated tests and Docker deployment.

Additional Professional Experience

Android / Flutter Developer | Freelancer.com | 2017 - Present

- Build mobile, desktop and web applications using Java, Kotlin, Dart/Flutter, APIs, databases, tests and deployment workflows.

AI Data Annotation and Automation | Independent / Contract | 2025 - 2026

- Built Python workflows using OCR, LLM integration, browser automation, recovery logic and quality controls.

AI Trainer / Data Quality Specialist | Freelance Platforms | 2016 - Present

- Prompt writing, response evaluation, fact-checking, STEM rubrics and bilingual Arabic/English quality review.

Education

M.Sc. Computer Science, Research Track | In progress

Arab Academy for Science, Technology and Maritime Transport (AASTMT).

B.Sc. Computer Science | Cairo University | 2024

Bachelor of Engineering, Mechanical Engineering | 2012

Higher Technology Institute (HTI), English Program.

Selected Training

Algorithms - Princeton University (online); Embedded Linux - Luxoft Egypt (2024 - 2025); Embedded Linux internship - Orange Digital Center with AMIT (2024); Embedded Systems - National Telecommunication Institute (2019).

Languages

Arabic: Native | English: Advanced, TOEIC Reading and Listening.